

# Reasoning in propositional logic theories: logical puzzles and exercises

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## Cards

Consider a deck of cards where each card has a letter in one face and one number on the other. One such deck is said to be “balanced” if any card that has a vowel in one face has an even number on the other side. Let  $M$  be a deck of 4 cards of each of which we are shown a face, as follows:

|                      |  |   |  |   |  |   |  |   |
|----------------------|--|---|--|---|--|---|--|---|
| card position        |  | 1 |  | 2 |  | 3 |  | 4 |
| writing on card face |  | A |  | 6 |  | K |  | 7 |

What is the minimum number of cards to turn over to verify that  $M$  is a balanced deck?

## Cards: solution

Consider a deck of cards where each card has a letter in one face and one number on the other. One such deck is said to be “balanced” if any card that has a vowel in one face has an even number on the other side. Let  $M$  be a deck of 4 cards of each of which we are shown a face, as follows:

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What is the minimum number of cards to turn over to verify that  $M$  is a balanced deck?

### Solution:

Formalize the rules capturing that a deck is balanced using a propositional theory  $F$  so that each deck is an interpretation; then determine the minimum number of letters propositional whose truth value must be determined for verify that  $M$  is a  $F$  pattern.

**Propositional letters:**  $v_1, v_2, v_3, v_4$  (vocali), and  $p_1, p_2, p_3, p_4$  (pari)

**Axioms of  $F$  and facts:**  $\{v_1 \rightarrow p_1, v_2 \rightarrow p_2, v_3 \rightarrow p_3, v_4 \rightarrow p_4\} \cup \{v_1, p_2, \neg v_3, \neg p_4\}$

**$M$  as an interpretation:**  $M(v_1) = M(p_2) = \text{True}$ ,  $M(v_3) = M(p_4) = \text{False}$

**Task:** determine the minimum number of letters you have to verify in order to check whether  $M$  can be extended to a model of  $F$

| $i$ | card | $v_i$ | $p_i$ | $v_i \rightarrow p_i$ |
|-----|------|-------|-------|-----------------------|
| 1   | A    | True  | ?     | ?                     |
| 2   | 6    | ?     | True  | True                  |
| 3   | K    | False | ?     | True                  |
| 4   | 7    | ?     | False | ?                     |

To complete the truth table we should know the value of  $p_1$  and  $v_4$ . Hence, we have to turn two cards: card 1 and card 4.

## Three perfect logicians at the pub

A perfect logician answers perfectly to every “boolean question”  $Q$ , with

- “yes” (if her/his knowledge implies  $Q$  true)
- “no” (if her/his knowledge implies  $Q$  false)
- “I do not know” (if her/his knowledge implies neither  $Q$  true, nor  $Q$  false)

as possible answers.

Three perfect logicians who have never met with each other enter into a pub and sit at the only table with free seats (exactly three seats). The waiter goes to the table and, since he is very busy, while cleaning the table, decides to ask collectively the following question: “Everybody at this table wants a beer?” The three logicians answer one by one, and after hearing the answers, the waiter brings exactly one beer at the table.

The question for you is: [Is the waiter a perfect logician?](#)

## Three perfect logicians at the pub: solution

Three perfect logicians who have never met with each other enter into a pub and sit at the only table with free seats (exactly three seats). The waiter goes to the table and, since he is very busy, while cleaning the table, decides to ask collectively the following question: “Everybody at this table wants a beer?” The three logicians answer one by one, and after hearing the answers, the waiter brings exactly one beer at the table. Is the waiter a perfect logician?

We formalize the puzzle in terms of a theory (set of axioms) representing the knowledge of the waiter. The alphabet of the theory is simply constituted by the propositional letters  $b_1, b_2, b_3, b_4$ , where  $b_i$  means that logician  $i$  wants a beer.

- The initial theory  $K_0$  is empty, because the waiter knows nothing.
- After the answer by logician  $i$  ( $i = 1, 2, 3$ ) the theory changes and the new theory will be denoted by  $K_i$ . Thus,  $K_3$  is the final theory.
- The waiter is a perfect logician if and only if, no matter what the three answers are,

$$K_3 \models (b_1 \wedge \neg b_2 \wedge \neg b_3) \vee (\neg b_1 \wedge b_2 \wedge \neg b_3) \vee (\neg b_1 \wedge \neg b_2 \wedge b_3)$$

## Three perfect logicians at the pub: solution

Initially  $K_0 = \{\}$ ; answers of the 1st logician:

- “yes”  $\rightsquigarrow$  impossible (logician 1 doesn't know about the others)
- “no”  $\rightsquigarrow$  all future answers will be “no”,  $K_1 = K_2 = K_3 = \{\neg b_1\}$  possible outcomes are 0, 1 or 2 beers and **the waiter should ask for more information**
- “I do not know”  $\rightsquigarrow K_1 = \{b_1\}$ ; answers of the 2nd logician:
  - ▶ “yes”  $\rightsquigarrow$  impossible (logician 2 doesn't know about logician 3)
  - ▶ “no” the third answer will be “no”,  $\rightsquigarrow K_2 = K_3 = \{b_1, \neg b_2\}$ , possible outcomes are 1 or 2 beers and **the waiter should ask for more information**
  - ▶ “I do not know”  $\rightsquigarrow K_2 = \{b_1, b_2\}$ , answers of the 3rd logician
    - ★ “yes”  $\rightsquigarrow K_3 = \{b_1, b_2, b_3\}$ , **the waiter should bring 3 beers**
    - ★ “no”  $\rightsquigarrow K_3 = \{b_1, b_2, \neg b_3\}$ , **the waiter should bring 2 beers**
    - ★ “I do not know”  $\rightsquigarrow$  impossible (logician 3 knows everything)

Conclusion: **the waiter is not a perfect logician**, because in all cases

$$K_3 \not\models (b_1 \wedge \neg b_2 \wedge \neg b_3) \vee (\neg b_1 \wedge b_2 \wedge \neg b_3) \vee (\neg b_1 \wedge \neg b_2 \wedge b_3)$$

## The 3 doors

### Problem

Kyle, Neal, and Grant find themselves trapped in a dark and cold dungeon. After a quick search the boys find three doors, the first one red, the second one blue, and the third one green.

Behind one of the doors is a path to freedom. Behind the other two doors, however, is an evil fire-breathing dragon. Opening a door to the dragon means certain death. On each door there is an inscription:

|                  |                      |                      |
|------------------|----------------------|----------------------|
| <b>freedom</b>   | <b>freedom</b>       | <b>freedom</b>       |
| <b>is behind</b> | <b>is not behind</b> | <b>is not behind</b> |
| <b>this door</b> | <b>this door</b>     | <b>the blue door</b> |

Given the fact that at LEAST ONE of the three statements on the three doors is true and at LEAST ONE of them is false, which door would lead the boys to safety?



# The 3 doors: solution

## Language

- $r$ : "freedom is behind the red door"
- $b$ : "freedom is behind the blue door"
- $g$ : "freedom is behind the green door"

## Axioms

- 1 "behind one of the door is a path to freedom, behind the other two doors is an evil dragon"  
 $(r \wedge \neg b \wedge \neg g) \vee (\neg r \wedge b \wedge \neg g) \vee (\neg r \wedge \neg b \wedge g)$
- 2 "at least one of the three statements is true"  
 $r \vee \neg b$
- 3 "at least one of the three statements is false"  
 $\neg r \vee b$

## The 3 doors: solution

### Axioms

- ①  $(r \wedge \neg b \wedge \neg g) \vee (\neg r \wedge b \wedge \neg g) \vee (\neg r \wedge \neg b \wedge g)$
- ②  $r \vee \neg b$
- ③  $\neg r \vee b$

### Solution

| $r$   | $b$   | $g$   | 1     | 2     | 3     | $2 \wedge 3$ | $1 \wedge 2 \wedge 3$ |
|-------|-------|-------|-------|-------|-------|--------------|-----------------------|
| False | False | False | False | True  | True  | True         | False                 |
| False | False | True  | True  | True  | True  | True         | True                  |
| False | True  | False | True  | False | True  | False        | False                 |
| False | True  | True  | False | False | True  | False        | False                 |
| True  | False | False | True  | True  | False | False        | False                 |
| True  | False | True  | False | True  | False | False        | False                 |
| True  | True  | False | False | True  | True  | True         | False                 |
| True  | True  | True  | False | True  | True  | True         | False                 |

Since  $\{1, 2, 3\} \models g$ , freedom is behind the **green door**!

### Problem

Provide the formalization in propositional logic of the graph coloring problem.

- Graph coloring problem: given a non-oriented graph with  $n > 0$  nodes, check whether the graph is  $k$ -colorable, i.e., whether we can associate one of the  $k > 0$  colors to each of its nodes in such a way that no pair of adjacent nodes have the same color.

# Graph coloring

## Problem

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## Solution:

Formalize the problem in terms of a propositional theory  $T_G$  whose axioms are satisfied if and only if the graph  $G$  is colorable.

# Graph coloring: propositional formalization

## Language of theory $T_G$

- For each  $1 \leq i \leq n$  and  $1 \leq c \leq k$ ,  $\text{color}_{ic}$  is a proposition, which intuitively means that *"the  $i$ -th node of  $G$  has the  $c$  color"*
- For each  $1 \leq i \neq j \leq n$ ,  $\text{edge}_{ij}$  is a proposition, which intuitively means that *"the  $i$ -th node is connected in  $G$  with the  $j$ -th node"*.

## Axioms of theory $T_G$

- for each  $1 \leq i \leq n$ ,  $\bigvee_{c=1}^k \text{color}_{ic}$   
*"each node has at least one color"*
- for each  $1 \leq i \leq n$  and  $1 \leq c \neq c' \leq k$ ,  $\text{color}_{ic} \rightarrow \neg \text{color}_{ic'}$   
*"every node has at most 1 color"*
- for each  $1 \leq i, j \leq n$  and  $1 \leq c \leq k$ ,  $\text{edge}_{ij} \rightarrow \neg(\text{color}_{ic} \wedge \text{color}_{jc})$   
*"adjacent nodes do not have the same color"*
- facts,  $\bigwedge \text{edge}_{ij}$   
*"all given edges are true"*

It can be shown that every solution to the graph coloring problem for  $G$  corresponds to a model of  $T_G$ , and, conversely, every model of  $T_G$  corresponds to a solution of the graph coloring problem for  $G$ .

# Sudoku

## Problem

Provide a formalization of the Sudoku game. Sudoku is a placement puzzle. The aim of the puzzle is to enter a numeral from 1 through 9 in each cell of a grid, here a  $9 \times 9$  grid made up of  $3 \times 3$  subgrids (called "regions"), starting with various numerals given in some cells (the "givens"). Each row, column and region must contain only one instance of each numeral. Its grid layout is like the one shown in the following schema

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   |   | 9 |   |   |   | 7 |   |   |
|   | 4 |   | 5 |   | 9 |   | 1 |   |
| 3 |   |   |   | 1 |   |   |   | 2 |
|   | 1 |   |   | 6 |   |   | 7 |   |
|   |   | 2 | 7 |   | 1 | 8 |   |   |
|   | 5 |   |   | 4 |   |   | 3 |   |
| 7 |   |   |   | 3 |   |   |   | 4 |
|   | 8 |   | 2 |   | 4 |   | 6 |   |
|   |   | 6 |   |   |   | 5 |   |   |

# Sudoku: solution

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|---|---|---|---|---|---|---|---|---|
|   |   | 9 |   |   |   | 7 |   |   |
|   | 4 |   | 5 |   | 9 |   | 1 |   |
| 3 |   |   |   | 1 |   |   |   | 2 |
|   | 1 |   |   | 6 |   |   | 7 |   |
|   | 2 | 7 |   |   | 1 | 8 |   |   |
|   | 5 |   |   | 4 |   |   | 3 |   |
| 7 |   |   |   | 3 |   |   |   | 4 |
|   | 8 |   | 2 |   | 4 |   | 6 |   |
|   |   | 6 |   |   |   | 5 |   |   |

We provide a formalization of the Sudoku game as a propositional theory, so that any truth assignment to the propositional variables that satisfy the axioms of the theory is a solution for the puzzle, and, conversely, every solution for the puzzle corresponds to a truth assignment to the propositional variables that satisfy the axioms of the theory.

## Sudoku: solution

### Language of theory $T_S$

For  $1 \leq n, r, c \leq 9$ , define the proposition

$$in(n, r, c)$$

which means that the number  $n$  has been inserted in the cross between row  $r$  and column  $c$ .



## Axioms of theory $T_S$

- ① "Every number from 1 to 9 appears in every row"

$$\bigwedge_{n=1}^9 \left( \bigwedge_{r=1}^9 \left( \bigvee_{c=1}^9 in(n, r, c) \right) \right)$$

- ② "Every number from 1 to 9 appears in every column"

$$\bigwedge_{n=1}^9 \left( \bigwedge_{c=1}^9 \left( \bigvee_{r=1}^9 in(n, r, c) \right) \right)$$

- ③ "Every numbers from 1 to 9 appears in every region (sub-grid)"

$$\text{for any } 0 \leq k, h \leq 2 \quad \bigwedge_{n=1}^9 \left( \bigvee_{r=1}^3 \left( \bigvee_{c=1}^3 in(n, 3 * k + r, 3 * h + c) \right) \right)$$

- ④ "No cell can contain more than one number"

$$\text{for any } 1 \leq n, n', c, r \leq 9 \text{ and } n \neq n' \quad in(n, r, c) \rightarrow \neg in(n', r, c)$$

- ⑤ "the givens" represented by appropriate  $in(n, r, c)$